

1. 在对流层内, 对流的空气近似经历绝热过程, 其绝热方程为  $\frac{p}{p_0} = \left(\frac{T}{T_0}\right)^{\frac{\gamma}{\gamma-1}} = K$  通过流体静力学平衡推导压强随高度的函数关系.

解: 流体静力学平衡方程:  $dp = -\rho g dz$  与绝热方程  $\frac{p}{p_0} = K$  联立可得

$$dp = -g \left(\frac{p}{K}\right)^{\frac{1}{\gamma}} dz \Rightarrow \frac{dp}{p^{\frac{1}{\gamma}}} = -g K^{-\frac{1}{\gamma}} dz \quad \text{取海平面处 } (p, p_0, z) = (p_0, p_0, z_0) \text{ 为边界条件.}$$

积分有  $\int_{p_0}^p \frac{dp}{p^{\frac{1}{\gamma}}} = \frac{p^{1-\frac{1}{\gamma}} - p_0^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} = -g K^{-\frac{1}{\gamma}} (z - z_0) \Rightarrow \text{代入 } K = \frac{p_0}{p_0} \text{ 后整理可得:}$

$$\left(\frac{p}{p_0}\right)^{1-\frac{1}{\gamma}} = 1 - (1-\frac{1}{\gamma}) g \frac{p_0}{p_0} (z - z_0) \quad \text{即 } p = p_0 \left[ 1 - (1-\frac{1}{\gamma}) \frac{p_0 g (z - z_0)}{p_0} \right]^{\frac{\gamma}{\gamma-1}}$$

2. Temperature  $T(z) = T_0 - L \cdot z$ , 推导地球大气压强随高度  $h$  的变化公式; 设  $L = 6.5 \text{ K/km}$ ,  $T_0 = 288 \text{ K}$ , 计算  $20 \text{ km}$  高处的地球大气压强。对比地球大气真实数据, 请说明产生差异的原因。

解: 流体静力学平衡方程  $dp = -\rho g dz = -\rho g dh$   
理想气体状态方程  $p = \frac{nRT}{V} = \rho \frac{R}{M} T = \rho R_d T$   $\Rightarrow dp = -\frac{\rho g}{R_d T} dh$  代入  $T = T_0 - Lh$  有

$$\frac{dp}{p} = -\frac{g dh}{R_d (T_0 - Lh)} \quad \text{取高度 } h=0 \text{ 处 } (p, T, h) = (p_0, T_0, 0) \text{ 为边界条件, 积分有}$$

$$\int_{p_0}^p \frac{dp}{p} = -\frac{g}{R_d} \int_0^h \frac{dh}{T_0 - Lh} \Rightarrow \ln \frac{p}{p_0} = \frac{g}{R_d L} \ln \frac{T_0 - Lh}{T_0} \Rightarrow p = p_0 \left(1 - \frac{Lh}{T_0}\right)^{\frac{g}{R_d L}}$$

$$\text{代入 } R_d = 287 \text{ J/(kg} \cdot \text{K)}, g = 9.8 \text{ m/s}^2, L = 6.5 \text{ K/km} = 6.5 \times 10^{-3} \text{ K/m}, h = 2 \times 10^4 \text{ m}, T_0 = 288 \text{ K}, p_0 = 1.01325 \times 10^5 \text{ Pa}$$

$$\text{可得 } p(h=20 \text{ km}) = 1.01325 \times 10^5 \times \left(1 - \frac{6.5 \times 20}{288}\right)^{\frac{9.8}{287 \times 6.5 \times 10^{-3}}} \approx 4.325 \times 10^3 \text{ Pa}$$

对比地球大气真实数据, 从 International standard Atmosphere (ISA) 查得  $20 \text{ km}$  处压强为  $5.53 \times 10^3 \text{ Pa}$ , 要大于我们求出的值。产生差异的主要差异是题设的温度递减率  $L = 6.5 \text{ K/km}$  在  $20 \text{ km}$  这样一个已进入平流层的高度不再适用。从 ISA 给出的  $T-h$  曲线图可看出, 在  $11 \text{ km}$  以下, 温度随高度增加线性递减, 而在  $11 \text{ km}$  以上, 即进入平流层后 ISA 近似采用了等温大气模型。

3. 假定地球大气中氧原子和氢原子的速度分布符合麦克斯韦-玻尔兹曼分布 (温度假设为  $288 \text{ K}$ ), 计算两种原子大于逃逸速度的比例。如果考虑到热层  $1000 \text{ km}$  处温度为  $2000 \text{ K}$ , 计算两种原子大于逃逸速度的比例。

$$\text{解: 逃逸速度 } v_{\text{esc}} \sqrt{\frac{2GM}{R}} \quad \text{故地表附近 } v_{\text{esc}} = \sqrt{\frac{2GM}{R_E}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6371 \times 10^3}} \approx 11.18 \text{ km/s}$$

$$\text{热层 } 1000 \text{ km 处: } v_{\text{esc}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{(6371 + 1000) \times 10^3}} \approx 10.39 \text{ km/s}$$

Maxwell-Boltzmann 分布给出速率分布在  $v \sim v+dv$  内的粒子数目

$$dN(v) = N_0 \left( \frac{m}{2\pi k_B T} \right)^{3/2} 4\pi v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) dv$$

故  $v > v_{esc}$  的粒子占的比例为  $\frac{N(v > v_{esc})}{N_0} = 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} \int_{v_{esc}}^{\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^2 dv$

$$= \frac{4}{\sqrt{\pi}} \int_{x_{esc}}^{\infty} e^{-x^2} x^2 dx \quad \left( \text{其中 } x = \left( \frac{m}{2k_B T} \right)^{1/2} v \right)$$

$$= -\frac{2}{\sqrt{\pi}} \int_{x_{esc}}^{\infty} x de^{-x^2} = -\frac{2}{\sqrt{\pi}} \left( x e^{-x^2} \Big|_{x_{esc}}^{\infty} - \int_{x_{esc}}^{\infty} e^{-x^2} dx \right) = \frac{2}{\sqrt{\pi}} x_{esc} e^{-x_{esc}^2} + \frac{2}{\sqrt{\pi}} \int_{x_{esc}}^{\infty} e^{-x^2} dx$$

代入误差函数  $\text{erf}(x) = \frac{1}{\sqrt{\pi}} \int_{-x}^x e^{-t^2} dt = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt = 1 - \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-t^2} dt$

有  $\frac{N(v > v_{esc})}{N_0} = \frac{2}{\sqrt{\pi}} x_{esc} e^{-x_{esc}^2} + 1 - \text{erf}(x_{esc}) = \frac{2}{\sqrt{\pi}} x_{esc} e^{-x_{esc}^2} + \text{erfc}(x_{esc})$

地表附近 ( $T=288\text{K}$ ): H:  $x_{H,esc} = \left( \frac{m_H}{2k_B T} \right)^{1/2} v_{esc} = \left( \frac{1.67 \times 10^{-27}}{2 \times 1.38 \times 10^{-23} \times 288} \right)^{1/2} \times 11.18 \times 10^3 \approx 5.124$

O:  $x_{O,esc} = \left( \frac{m_O}{2k_B T} \right)^{1/2} v_{esc} = \left( \frac{16 \times 1.67 \times 10^{-27}}{2 \times 1.38 \times 10^{-23} \times 288} \right)^{1/2} \times 11.18 \times 10^3 \approx 20.498$

热层 1000km 处 ( $T=2000\text{K}$ ): H:  $x_{H,esc} = \left( \frac{1.67 \times 10^{-27}}{2 \times 1.38 \times 10^{-23} \times 2000} \right)^{1/2} \times 10.39 \times 10^3 \approx 1.807$

O:  $x_{O,esc} = \left( \frac{16 \times 1.67 \times 10^{-27}}{2 \times 1.38 \times 10^{-23} \times 2000} \right)^{1/2} \times 10.39 \times 10^3 \approx 7.229$

代入各自的  $x_{esc}$ , 其中  $\text{erf}(x_{esc})$  可从 Python ~~math~~ `scipy.special.erfc` 函数计算得到, 可得各自比例为:

地表附近 ( $T=288\text{K}$ )

$$\begin{cases} \text{H: } \frac{N(v > v_{esc})}{N_0} = \frac{2}{\sqrt{\pi}} \times 5.124 \times e^{-5.124^2} + 1 - \text{erf}(5.124) \approx 2.288 \times 10^{-11} + 4.279 \times 10^{-13} \\ \text{O: } \frac{N(v > v_{esc})}{N_0} = \frac{2}{\sqrt{\pi}} \times 20.498 \times e^{-20.498^2} + 1 - \text{erf}(20.498) \approx 2.331 \times 10^{-11} + 7.718 \times 10^{-182} \end{cases}$$

热层 1000km 处 ( $T=2000\text{K}$ )

$$\begin{cases} \text{H: } \frac{N(v > v_{esc})}{N_0} = \frac{2}{\sqrt{\pi}} \times 1.807 \times e^{-1.807^2} + 1 - \text{erf}(1.807) \approx 7.786 \times 10^{-2} + 1.060 \times 10^{-2} \\ \text{O: } \frac{N(v > v_{esc})}{N_0} = \frac{2}{\sqrt{\pi}} \times 7.229 \times e^{-7.229^2} + 1 - \text{erf}(7.229) \approx 1.644 \times 10^{-22} + 0.156 \times 10^{-23} \end{cases}$$

可见热层附近由于  $T$  高, 热运动速度大, 逃逸比例要远大于地表附近.

5. 对于一维大气分布, 仅考虑瑞利散射, 计算不同天顶距 ( $z < 60^\circ$ ), V波段 ( $545 \pm 85 \text{ nm}$ ) 的平均大气透过率, 画出透过率随  $z$  变化的图片, 如何考虑  $z > 90^\circ$  的修正。

解: 瑞利散射的光学厚度为  $\tau_\lambda(z=0) = 0.00877 \lambda^{-4.05}$  (其中  $\lambda$  的单位为  $\mu\text{m}$ )

故 V波段 ( $0.46 \mu\text{m} \sim 0.63 \mu\text{m}$ ) 平均光学厚度为

$$\tau_v(z=0) = \frac{0.00877 \int_{0.46}^{0.63} \lambda^{-4.05} d\lambda}{0.63 - 0.46} = \frac{0.00877 \cdot \frac{1}{-3.05} \left( \frac{1}{0.46^{3.05}} - \frac{1}{0.63^{3.05}} \right)}{0.63 - 0.46} \approx 0.1114$$

相对大气质量 (大气质量数) 的定义为光线自  $z > 0$  处入射时与自天顶 ( $z=0$ ) 入射时的光学厚度之比:

$$m(z) = \frac{\tau_v(z)}{\tau_v(z=0)} = \frac{\int_0^\infty k(s) ds}{\int_0^\infty k(h) dh}$$

则有  $\tau_v(z) = \tau_v(z=0) \cdot m(z)$

由朗伯-比不定律有: V波段平均大气透过率  $T_v(z) = \frac{I}{I_0} = e^{-\tau_v(z)} = e^{-\tau_v(z=0) \cdot m(z)} = e^{-0.1114 m(z)}$

当  $z < 60^\circ$  时, 考虑 ~~平面平行层~~ 大气分布

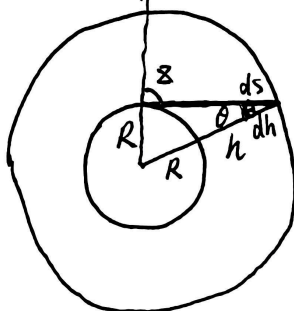


$$m(z) = \frac{\int_0^\infty k(h) \sec z dh}{\int_0^\infty k(h) dh} = \sec z$$

故  $T_v(z) = e^{-0.1114 \sec z}$

当  $z \geq 60^\circ$  时, 需考虑一维球对称大气分布

$$ds = \frac{dh}{\cos \theta} = \frac{dh}{\sqrt{1 - \left(\frac{R}{R+h} \sin z\right)^2}}$$



正弦定理  $\frac{\sin \theta}{R} = \frac{\sin(180^\circ - z)}{R+h}$

$$\Rightarrow \sin \theta = \frac{R}{R+h} \sin z$$

$$\Rightarrow \cos \theta = \sqrt{1 - \left(\frac{R}{R+h} \sin z\right)^2}$$

故  $m(z) = \frac{\int_0^\infty k(h) \frac{1}{\sqrt{1 - \left(\frac{R}{R+h} \sin z\right)^2}} dh}{\int_0^\infty k(h) dh}$   $T_v(z) = e^{-0.1114 m(z)}$

上式需通过数值积分计算, 较为复杂, 可采用 Kasten and Young (1989) 经验公式计算:

$$m(z) = \frac{1}{\cos z + 0.50572 \cdot (96.07995 - z)^{-1.6364}} \quad (z \text{ 以度为单位})$$

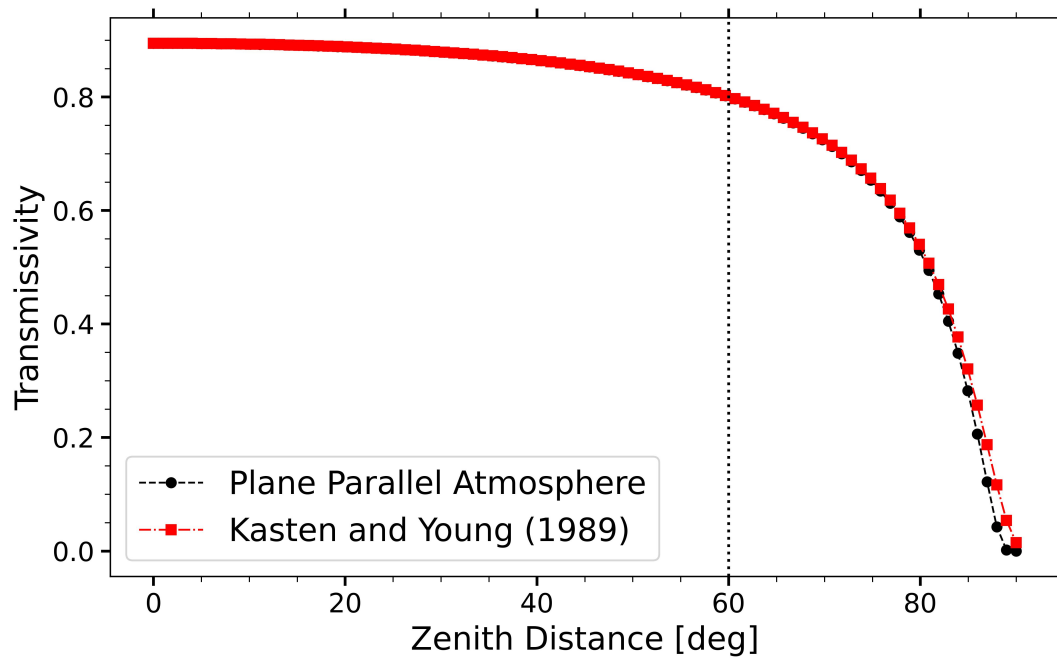
透过率随  $z$  变化的曲线  $T_v(z) = e^{-0.1114 m(z)}$ , 其中  $m(z) = \begin{cases} \sec(z) & \text{平面平行层大气} \\ \text{Kasten经验公式} & \end{cases}$

绘制如下:

```

transmissivity.py ×
1 import numpy as np
2 import matplotlib as mpl
3 from matplotlib import pyplot as plt
4
5 zlist = np.linspace(start=0, stop=90, num=90)
6 def Tp(z): 1个用法
7     z_rad = np.radians(z)
8     m = 1/np.cos(z_rad)
9     return np.exp(-0.1114*m)
10 def TK(z): 1个用法
11     z_rad = np.radians(z)
12     m = 1/(np.cos(z_rad)+0.50572*np.power(96.07995-z,-1.6364))
13     return np.exp(-0.1114 * m)
14 fig, ax = plt.subplots(figsize=(8,8*0.618), constrained_layout=True)
15 ax.plot(*args: zlist,[Tp(z) for z in zlist],c='black',ls='--',marker='o',lw=1,ms=5,label='Plane Parallel Atmosphere')
16 ax.plot(*args: zlist,[TK(z) for z in zlist],c='red',ls='-.',marker='s',lw=1,ms=5,label='Kasten and Young (1989)')
17 ax.axvline(x=60,c='black',ls=':')
18 ax.set_xlabel(xlabel: 'Zenith Distance [deg]',fontsize='xx-large')
19 ax.set_ylabel(ylabel: 'Transmissivity',fontsize='xx-large')
20 ax.xaxis.set_minor_locator(mpl.ticker.MultipleLocator(5))
21 ax.yaxis.set_minor_locator(mpl.ticker.MultipleLocator(0.05))
22 ax.legend(loc='best',fontsize='xx-large')
23 plt.tick_params(right='on', top='on', which='both')
24 ax.tick_params(which='major',length=5,width=1.5,labelsize='x-large')
25 ax.tick_params(which='minor',length=3)
26 plt.show()
27 fig.savefig(fname: 'Transmissivity.png', format='png', dpi=600, bbox_inches='tight', pad_inches=0.2)

```





6. 利用 Henyey-Greenstein 函数拟合瑞利散射, 得到对应的拟合参数的  $g$  的数值. 考虑是否有更好的函数形式来拟合瑞利散射.

解: Henyey-Greenstein 函数 (归一化到  $4\pi$ ): 
$$P_{HG}(\theta, g) = \frac{1-g^2}{(1+g^2-2g\cos\theta)^{3/2}}$$

瑞利散射相函数: 
$$P_R(\theta) = \frac{3}{4}(1+\cos^2\theta)$$

可见瑞利散射是轴对称且前向、后向对称的, 但对 Henyey-Greenstein 函数而言,  $g=0$  时各向同性,  $g>0$  时前向散射主导,  $g<0$  时后向散射主导, 都不符合瑞利散射相函数的特征, 因此直接拟合可能效果不佳.

~~Haltrin et al (2002)~~

Vladimir I Haltrin (2002) 在其文中的第 (6) 式给出了最初由 Kattawar (1975) 提出的

Two-Term Henyey-Greenstein Phase function: 
$$P_{THG}(\theta, \alpha, g_1, g_2) = \alpha P_{HG}(\theta, g_1) + (1-\alpha) P_{HG}(\theta, -g_2)$$

其中  $0 \leq \alpha, g_1, g_2 \leq 1$ , 对于瑞利散射的拟合, 考虑到其前/后向散射对称性, 可构造如下 THG 相函数来作拟合: 
$$P_{THG}(\theta, g) = \frac{1}{2}(P_{HG}(\theta, g) + P_{HG}(\theta, -g))$$

利用 python 库函数 `scipy.optimize.curve_fit` 拟合, 结果如下:

$P_{HG}$  拟合:  $g = 0.005371 \pm 0.013937$  (拟合效果不佳, 误差较大)

$P_{THG}$  拟合:  $g = 0.300743 \pm 0.001484$

```

transmissivity.py x Henyey-Greenstein.py x
1 import numpy as np
2 from scipy.optimize import curve_fit
3 import matplotlib.pyplot as plt
4
5 def rayleigh(cos_theta): 1用法
6     return (3/4)*(1+cos_theta**2)
7
8 def HG(cos_theta,g):
9     return (1-g**2)/((1+g**2-2*g*cos_theta)**(3/2))
10
11 def TTHG(cos_theta,g): 2用法
12     return (HG(cos_theta,g)+HG(cos_theta,-g))/2
13
14 theta = np.linspace(start=0,2*np.pi,num=100)
15 cos_theta = np.array([np.cos(th) for th in theta])
16 P_rayleigh = rayleigh(cos_theta)
17
18 g0 = 0.1
19 poptHG, pcovHG = curve_fit(HG, cos_theta, P_rayleigh, p0=[g0])
20 g_HG = poptHG[0]
21 print("HG fitting: g={:.6f} $\pm$ {:.6f}".format(*args: g_HG, np.sqrt(pcovHG[0][0])))
22 poptTTHG, pcovTTHG = curve_fit(TTHG, cos_theta, P_rayleigh, p0=[g0])
23 g_TTHG = poptTTHG[0]
24 print("TTHG fitting: g={:.6f} $\pm$ {:.6f}".format(*args: g_TTHG, np.sqrt(pcovTTHG[0][0])))
25 P_HG = HG(cos_theta, g_HG)
26 P_TTHG = TTHG(cos_theta, g_TTHG)
27 fig, ax = plt.subplots(figsize=(12,12),constrained_layout=True,subplot_kw={'projection': 'polar'})
28 ax.plot(*args: theta,P_rayleigh, c='black',ls='-',marker='.',label='Rayleigh Scattering Phase Function')
29 ax.plot(*args: theta,P_HG, c='red',ls='--',marker='s',label='HG Fitting: '+\
30         "g={:.6f} $\pm$ {:.6f}".format(*args: g_HG,np.sqrt(pcovHG[0][0])))
31 ax.plot(*args: theta,P_TTHG, c='lime',ls='--',marker='o',label='TTHG Fitting: '+\
32         "g={:.6f} $\pm$ {:.6f}".format(*args: g_TTHG,np.sqrt(pcovTTHG[0][0])))
33 ax.legend(loc='upper center',fontsize='xx-large')
34 ax.tick_params(which='major',labelsize='x-large')
35 plt.show()
36 fig.savefig(fname='Henyey-Greenstein.png', format='png', dpi=600, bbox_inches='tight', pad_inches=0.2)

```

